Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

Matthew Hockert*

April 10, 2026

Abstract

*University of Minnesota. Email: hocke043@umn.edu

1 Introduction

2 Institutional Background

2.1 Ohio Municipal Income Tax

2.2 Pennsylvania Earned Income Tax

3 Data

3.1 LODES Origin-Destination Data

3.2 Municipal Tax Rate Panels

3.3 Municipal Financial Reports

3.4 Outcome Variable: share_live_in

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Pennsylvania (2002–2019)					
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Panel B: Ohio (2002–2019)					
share_live_in	0.119	0.099	0.085		
Municipal tax rate (%)	1.222	1.500	0.821	0	3.000
Panel dimensions					
PA municipality × years	24,528 (1,383 municipalities)				
OH municipality × years	7,718 (430 municipalities)				

Notes: Municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. OH municipalities are Census places; PA municipalities are county subdivisions.

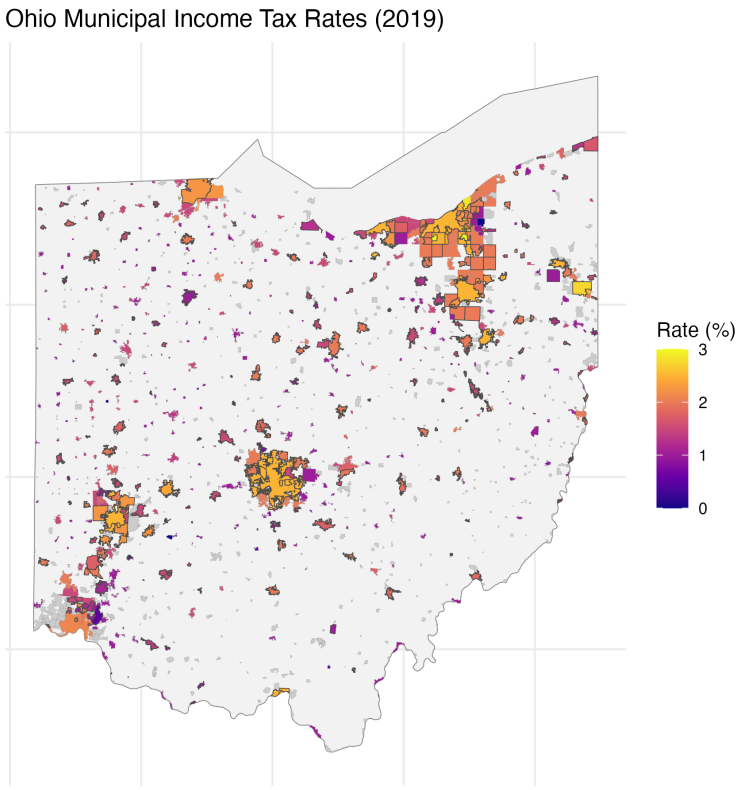


Figure 1: Ohio Municipal Income Tax Rates (2019). Bold outline = municipalities that raised rates during the sample period. Grey fill = municipalities with no municipal income tax.

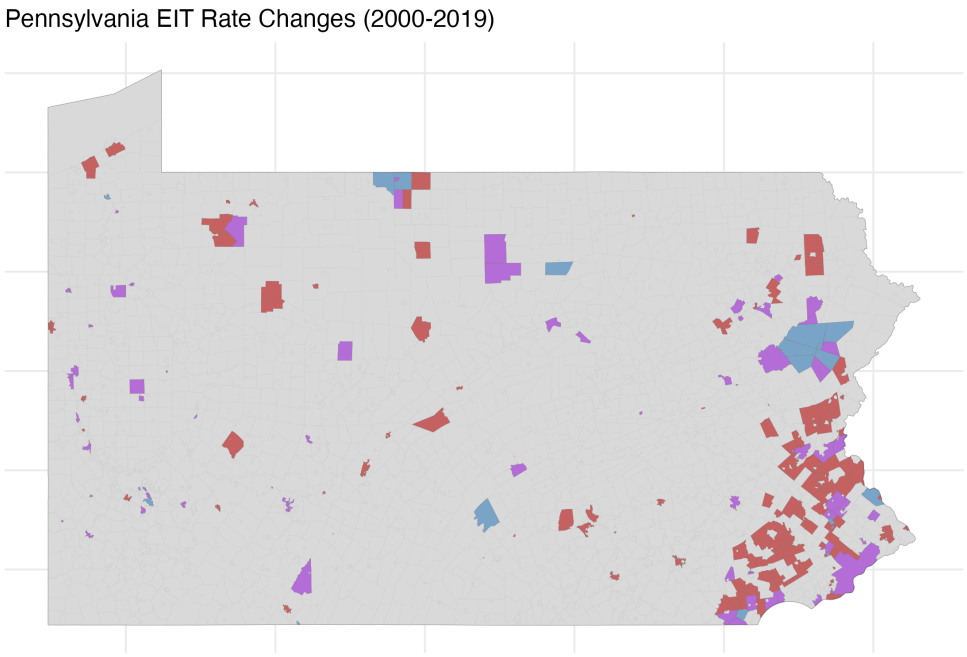


Figure 2: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

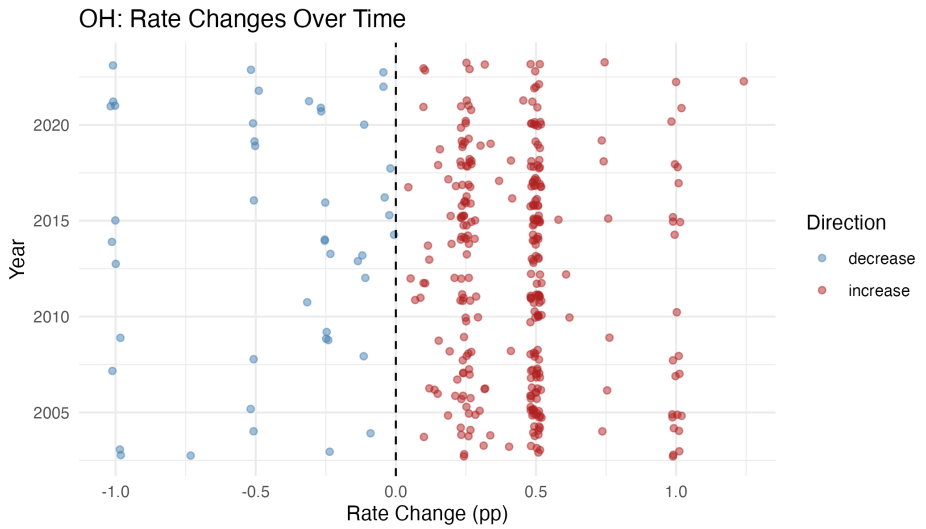


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

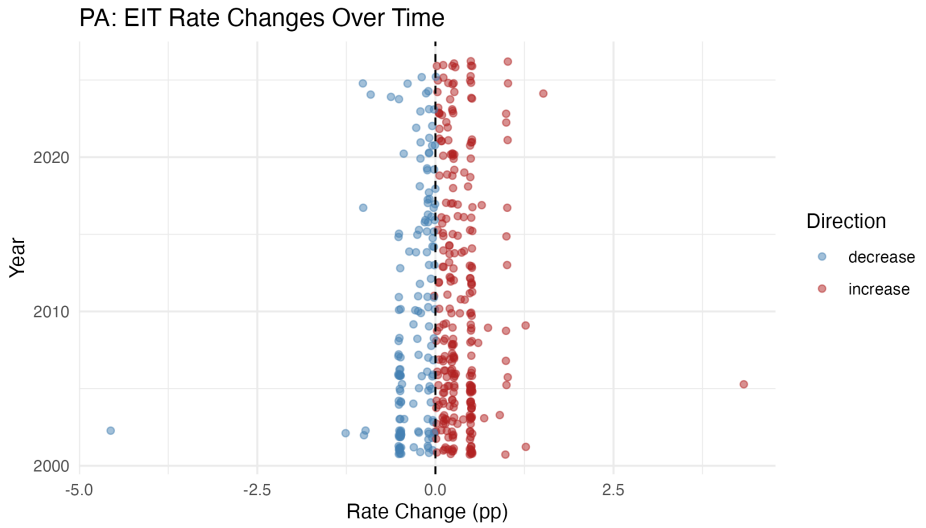


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

4 Empirical Strategy

Our empirical strategy proceeds in two steps. First, we estimate difference-in-differences models comparing municipalities that changed their tax rates to those that never changed. Because tax changes occur at different times across municipalities, we use the Sun and Abraham (2021) interaction-weighted estimator, which avoids the bias from heterogeneous treatment timing that can affect standard two-way fixed effects. We estimate rate increases and decreases separately, each against a clean never-changed control group, and then jointly to test for asymmetry. Second, we estimate dose-response models testing whether the magnitude of rate changes predicts additional sorting beyond the binary treatment effect.

4.1 Event Study Specification

Our baseline specification is a Sun and Abraham (2021) event study:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{1}$$

where Y_{it} is share_live_in for municipality i in year t; α_i and γ_t are municipality and year fixed

effects; D_{it}^{ℓ} is an indicator for municipality i being ℓ years from its first rate change; $\mathbf{X}_{it}$ is a vector of fiscal controls (log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid); and μ_ℓ is the treatment effect at relative time ℓ, estimated via the interaction-weighted approach of Sun and Abraham (2021). We normalize to $\ell = -1$ and cluster standard errors at the municipality level. Pre-treatment coefficients ($\ell < -1$) test the parallel trends assumption. Fiscal controls are available for Pennsylvania from the DCED Annual Financial Reports; we report specifications with and without controls.

We estimate this event study three times for each state:

1. **Increasers vs. never-changed.** Treated municipalities are those that raised their tax rate. The control group is municipalities that never changed their rate. Municipalities that only decreased rates are excluded from the sample entirely.

2. **Decreasers vs. never-changed.** The mirror test: treated municipalities are those that lowered their rate. Increasers are excluded.

3. **Asymmetry test.** Both increase and decrease indicators enter the same model, with the never-changed group as the common control:

$$Y_{it} = \alpha_i + \gamma_t + \delta^{+} \cdot \text{Post}_{it}^{+} + \delta^{-} \cdot \text{Post}_{it}^{-} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{2}$$

where $\text{Post}_{it}^{+} = 1$ for municipalities that raised rates, in all years after their first increase, and analogously Post_{it}^{-} for decreases. Comparing δ^{+} and δ^{-} tests whether sorting responses are symmetric.

Control groups. In Pennsylvania, the control group consists of 1,211 municipalities that never changed their EIT rate during the panel. Because all PA municipalities levy an EIT, identification comes from rate changes among universally taxed municipalities.

In Ohio, we report results with two control groups. The first is 104 municipalities that never levied an income tax, providing a clean extensive-margin comparison. The second

is 230 municipalities that maintained a stable tax rate (including taxed cities that never adjusted). The no-tax control group produces cleaner pre-trends, likely because stable-rate taxed cities share unobserved urban characteristics with treated cities.

4.2 Dose-Response

Among municipalities that raised rates, we test whether rate magnitude predicts additional sorting beyond a common post-treatment shift:

$$Y_{it} = \alpha_i + \gamma_t + \delta \cdot \text{Post}_{it} + \beta \cdot (\text{Rate}_{it} \times \text{Post}_{it}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \tag{3}$$

where $\text{Post}_{it} = 1$ in all years after municipality i's first rate increase. The sample is restricted to municipalities that raised rates. The coefficient δ captures common post-treatment shifts (e.g., fiscal stress that prompted the rate increase), while β isolates the tax channel. If fiscal distress alone drove the result, rate magnitude should not matter conditional on being a treated city.

To test for asymmetry in the dose-response, we decompose cumulative rate changes from baseline on the full sample:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ \cdot R^+_{it} + \beta^- \cdot R^-_{it} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \tag{4}$$

where $R^+_{it} = \max(\text{rate}_{it} - \text{rate}_{i0},\ 0)$ is the cumulative rate increase from baseline (zero if rate has not risen) and $R^-_{it} = |\min(\text{rate}_{it} - \text{rate}_{i0},\ 0)|$ is the absolute value of the cumulative rate decrease. If sorting is symmetric, β^+ and β^- should be equal in magnitude with opposite signs. Asymmetry — specifically $|\beta^+| > |\beta^-|$ — would indicate that residential location decisions are sticky: tax increases push residents out, but tax decreases do not pull them back.

5 Results

5.1 Main Results

Table 2: Binary DiD: Effect of Rate Changes on share_live_in (PA)

	(1)	(2)	(3)
	Increases vs NC	Decreases vs NC	Both vs NC
Sun-Abraham ATT	$-0.0053^{\cdot}$	0.0048	
	(0.0029)	(0.0057)	
increase × post			-0.0074^{*}
			(0.0037)
decrease × post			−0.0002
			(0.0060)
Treated	157 cities	63 cities	220 cities
Control	1,211 NC	1,211 NC	1,211 NC
N	24,522	24,522	24,522
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Dependent variable is share_live_in. NC = never-changed (municipalities that never adjusted their EIT rate). Columns (1)–(2): Sun-Abraham (2021) interaction-weighted ATT; cohort defined by year of first increase (decrease). Column (1) excludes decreasers; column (2) excludes increasers. Column (3): both in one model. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

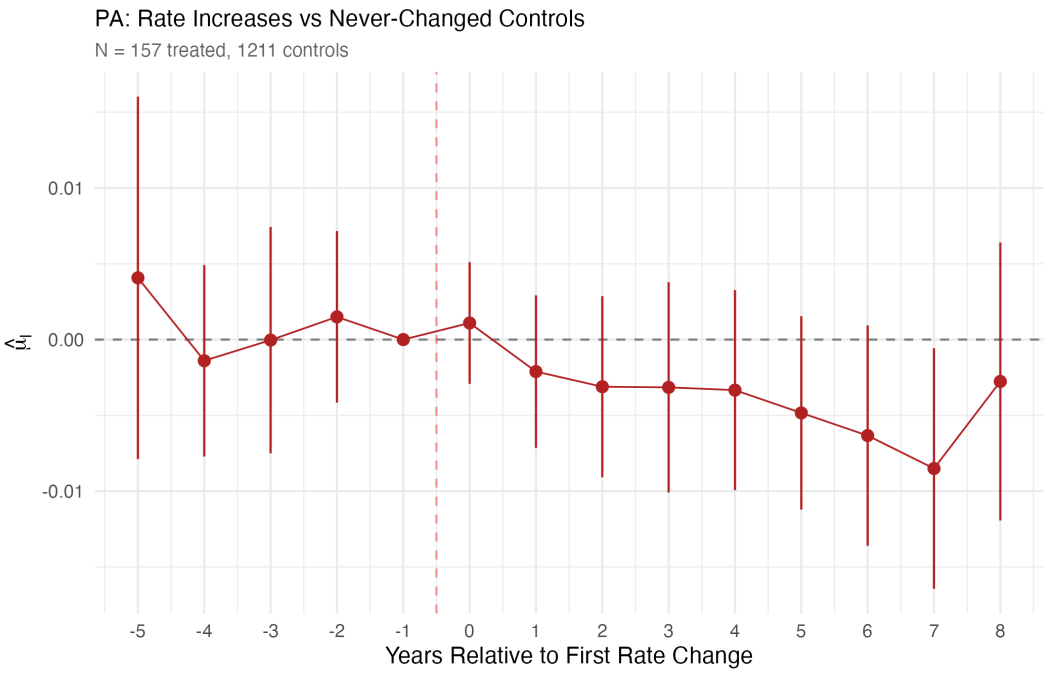


Figure 5: PA: Sun-Abraham Event Study — Rate Increases vs Never-Changed Controls. Cohort = year of first EIT increase. 157 treated municipalities, 1,211 never-changed controls. Pre-trends are flat; effect emerges gradually after treatment.

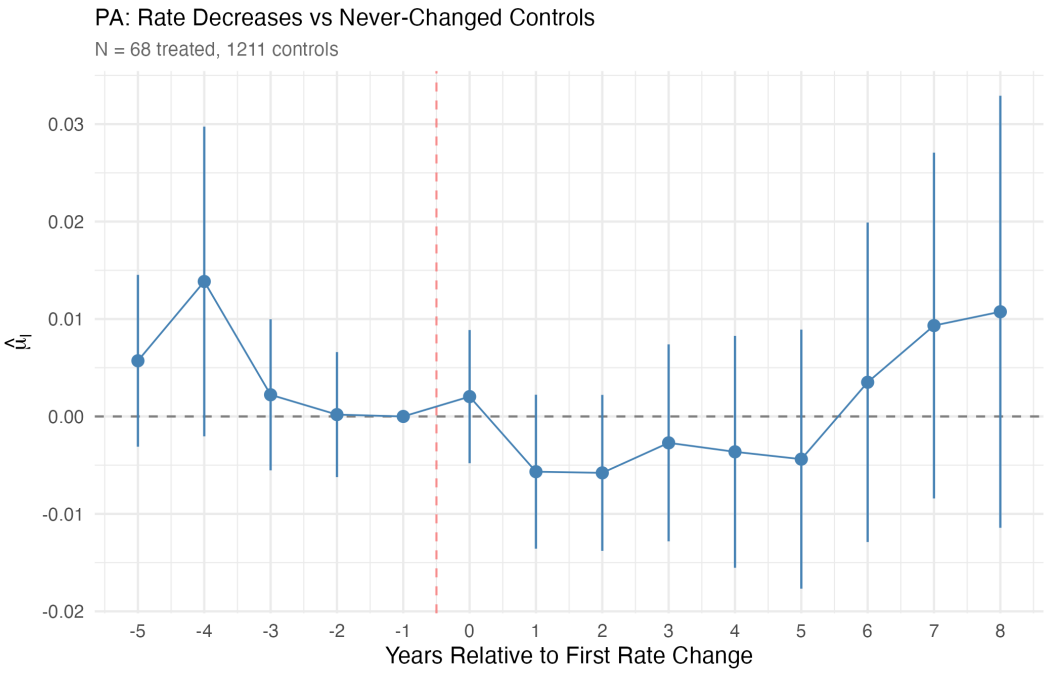


Figure 6: PA: Sun-Abraham Event Study — Rate Decreases vs Never-Changed Controls. Cohort = year of first EIT decrease. 63 treated municipalities, 1,211 never-changed controls. No evidence that rate decreases increase the locally-resident share.

Table 3: Binary DiD: Effect of Rate Changes on share_live_in (OH, No-Tax Controls)

	(1)	(2)	(3)
	Increases vs No-Tax	Decreases vs No-Tax	Both vs No-Tax
Sun-Abraham ATT	-0.0076^{**}	-0.0042	
	(0.0025)	(0.0064)	
increase × post			-0.0055^{**}
			(0.0020)
decrease × post			-0.0080
			(0.0126)
Treated	175 cities	25 cities	200 cities
Control	104 no-tax	104 no-tax	104 no-tax
N	5,081	5,081	5,081
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Control group = municipalities that never levied an income tax (unincorporated townships). Column (1) excludes decreasers from sample. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

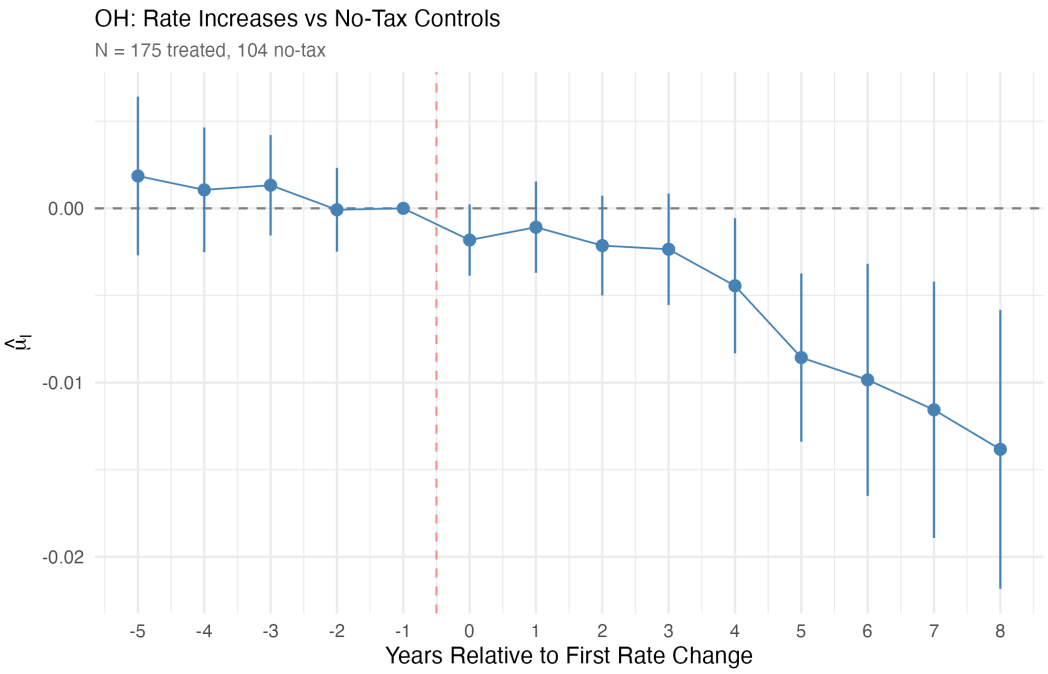


Figure 7: OH: Sun-Abraham Event Study — Rate Increases vs No-Tax Controls. 175 treated cities, 104 no-tax controls. Clean pre-trends; gradual decline through t+8.

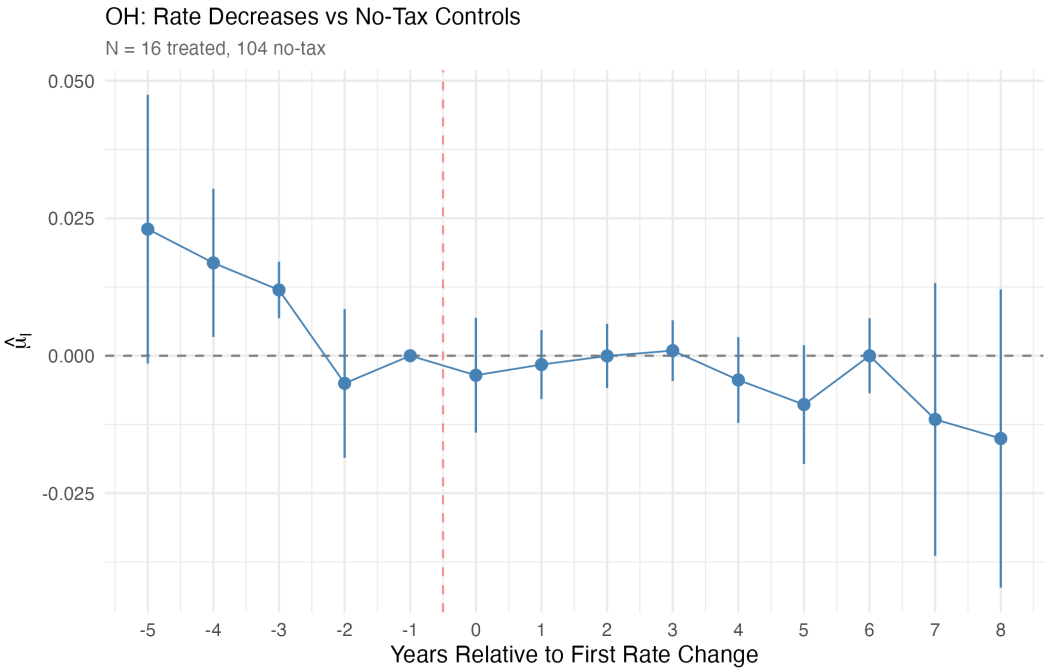


Figure 8: OH: Sun-Abraham Event Study — Rate Decreases vs No-Tax Controls. 25 treated cities, 104 no-tax controls. Wide confidence intervals; no clear pattern.

Table 4: Binary DiD: Effect of Rate Changes on share_live_in (OH, Never-Changed Controls)

	(1)	(2)	(3)
	Increases vs NC	Decreases vs NC	Both vs NC
Sun-Abraham ATT	−0.0002	0.0021	
	(0.0021)	(0.0063)	
increase × post			0.0004
			(0.0018)
decrease × post			−0.0054
			(0.0126)
Treated	175 cities	25 cities	200 cities
Control	230 NC	230 NC	230 NC
N	5,859	5,859	5,859
Fixed Effects		Municipality, Year	
Clusters		Municipality	

Notes: Control group = municipalities that never changed their tax rate (includes stable-rate taxed cities). Null results suggest that never-changed taxed cities are a poor control group, likely because they share the same urban characteristics as treated cities. The no-tax control group (Table 3) provides cleaner identification. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

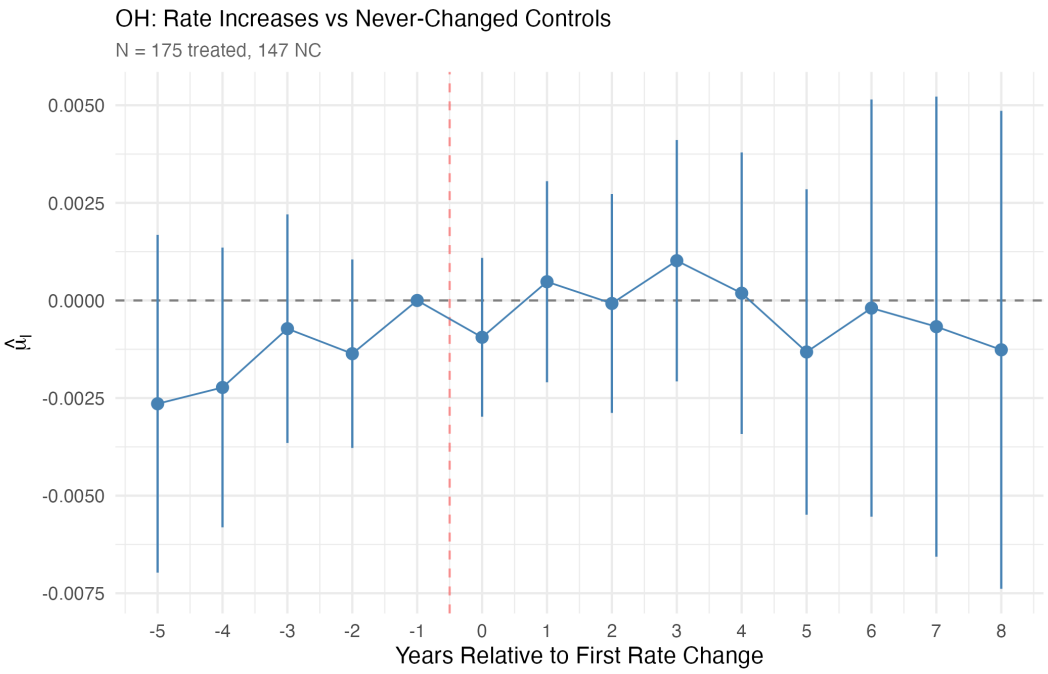


Figure 9: OH: Sun-Abraham Event Study — Rate Increases vs Never-Changed Controls. 175 treated cities, 230 never-changed controls (including stable-rate taxed cities). Pre-trends show some drift; effect is attenuated relative to the no-tax control group.

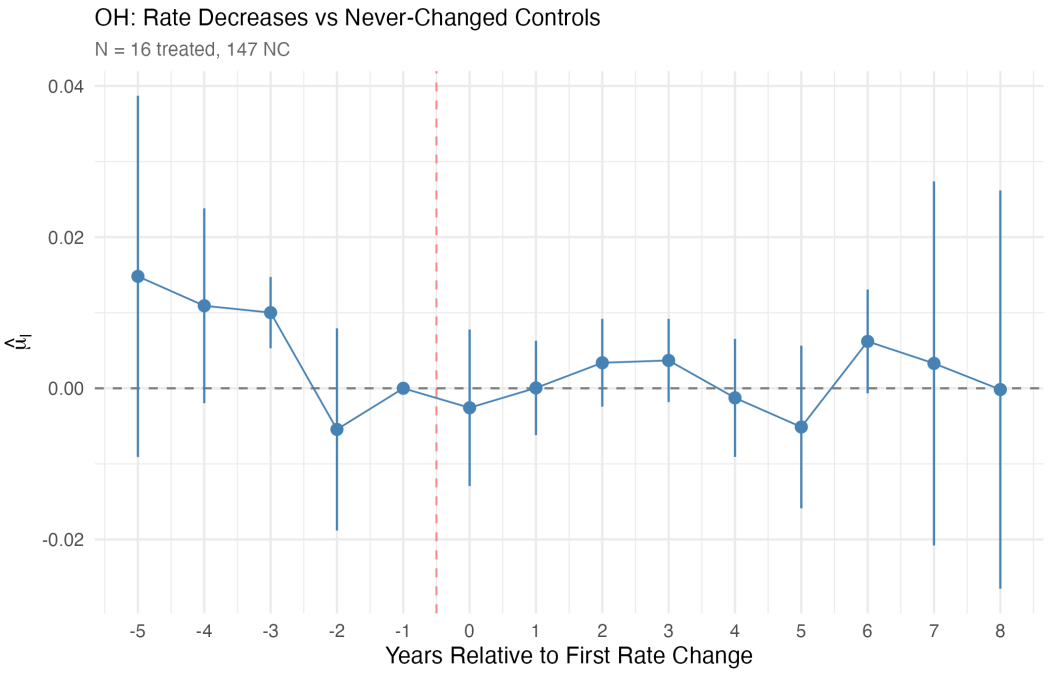


Figure 10: OH: Sun-Abraham Event Study — Rate Decreases vs Never-Changed Controls.

5.2 Dose-Response

Table 5: Dose-Response: Rate Magnitude and Asymmetry

	Pennsylvania			Ohio		
	(1)	(2)	(3)	(4)	(5)	(6)
	Post Only	Post + Dose	Split ±	Post Only	Post + Dose	Split ±
post	$-0.0074^{\cdot}$	0.0022		0.0005	−0.0036	
	(0.0040)	(0.0063)		(0.0017)	(0.0066)	
rate × post		$-0.0099^{\cdot}$			0.2155	
		(0.0052)			(0.3395)	
Δrate^{+}			-0.0176^{*}			−0.1091
			(0.0069)			(0.3788)
$\lvert\Delta\text{rate}^{-}\rvert$			0.0012			−1.155
			(0.0040)			(1.353)
N	2,741	2,741	24,522	3,150	3,150	7,718
Sample	Treated	Treated	All	Treated	Treated	All
Fixed Effects			Municipality, Year			
Clusters			Municipality			

Notes: Columns (1)–(2), (4)–(5): sample restricted to municipalities that raised rates; post = $\mathbb{1}[t \geq g_i]$. Column (2): rate × post tests whether rate magnitude predicts additional sorting beyond the common post-treatment shift. Columns (3), (6): split dose on full sample; Δrate^{+} = cumulative increase from baseline, $\lvert\Delta\text{rate}^{-}\rvert$ = cumulative decrease (absolute value). PA split dose: rate increases reduce locally-resident share (-0.018^{*} per pp); rate decreases have no effect (asymmetry). OH rates are in 0–3 scale (not percentage points), explaining the different coefficient magnitudes. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 6: Dose-Response: de Chaisemartin–D'Haultfoeuille (2024) Estimator

	Pennsylvania			Ohio	
	(1)	(2)	(3)	(4)	(5)
	Binary	Continuous	Continuous	Binary	Continuous
	Inc vs NC	Inc vs NC	Dec vs NC	Inc vs NC	Inc vs NC
Avg Total Effect	−0.0036	−0.0078	−0.0077	−0.0015	0.0446
	(0.0025)	(0.0069)	(0.0090)	(0.0014)	(0.3132)
Joint effects p	0.666	0.396	0.014	0.206	0.500
Joint placebo p	0.857	0.904	0.408	0.627	0.147
Switchers	109	120	60	165	174
Treatment	Binary	EIT rate	EIT rate	Binary	Muni rate
Control	Never-changed			Never-changed	

Notes: De Chaisemartin and D'Haultfoeuille (2024) interaction-weighted estimator. Columns (1), (4): binary treatment (1 after first rate increase). Columns (2), (5): continuous treatment (actual tax rate) with `continuous(1)` linear polynomial in baseline treatment. Column (3): continuous treatment among municipalities that decreased rates; the negative ATE indicates that rate decreases do not reverse residential sorting (asymmetry). Increasers and decreasers are estimated on separate samples, each against the same never-changed control group. Average Total Effect is the cumulative per-unit treatment effect averaged over ∼4.3 post-treatment periods. Event study dynamics shown in Figures 11–13. Analytical standard errors clustered at the municipality level. OH continuous estimates are imprecise due to the 0–3 rate scale. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

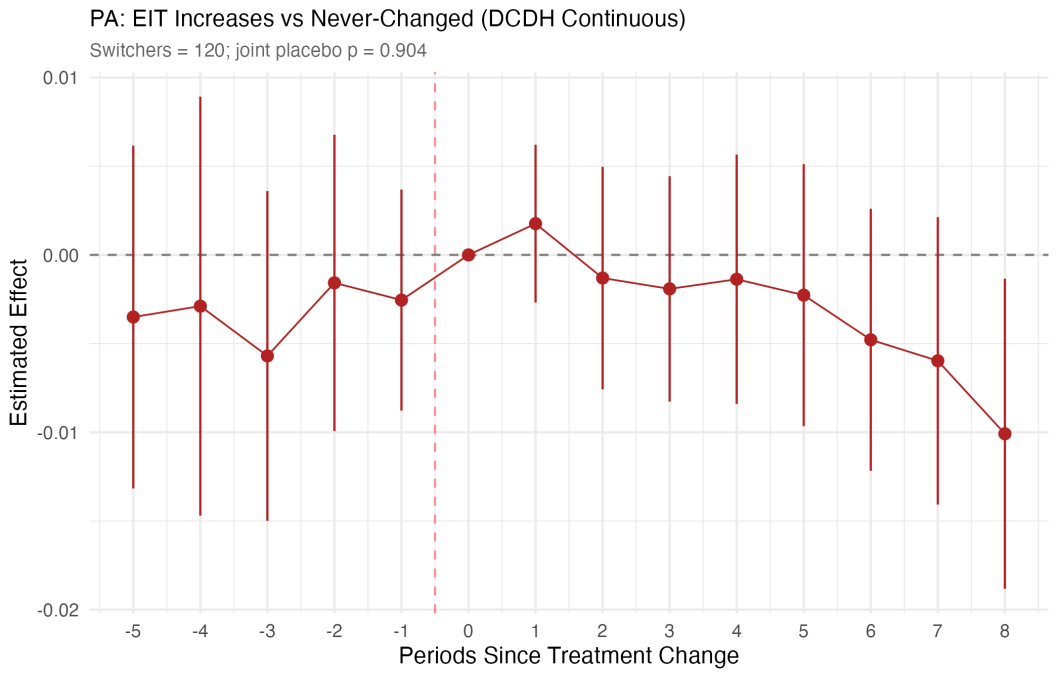


Figure 11: PA: DCDH Event Study — Continuous EIT Rate, Increasers vs Never-Changed. 120 switchers. Pre-trends clean (joint placebo $p = 0.90$). Effect builds gradually and reaches -0.011 ($p < 0.001$) at $\ell = 8$.

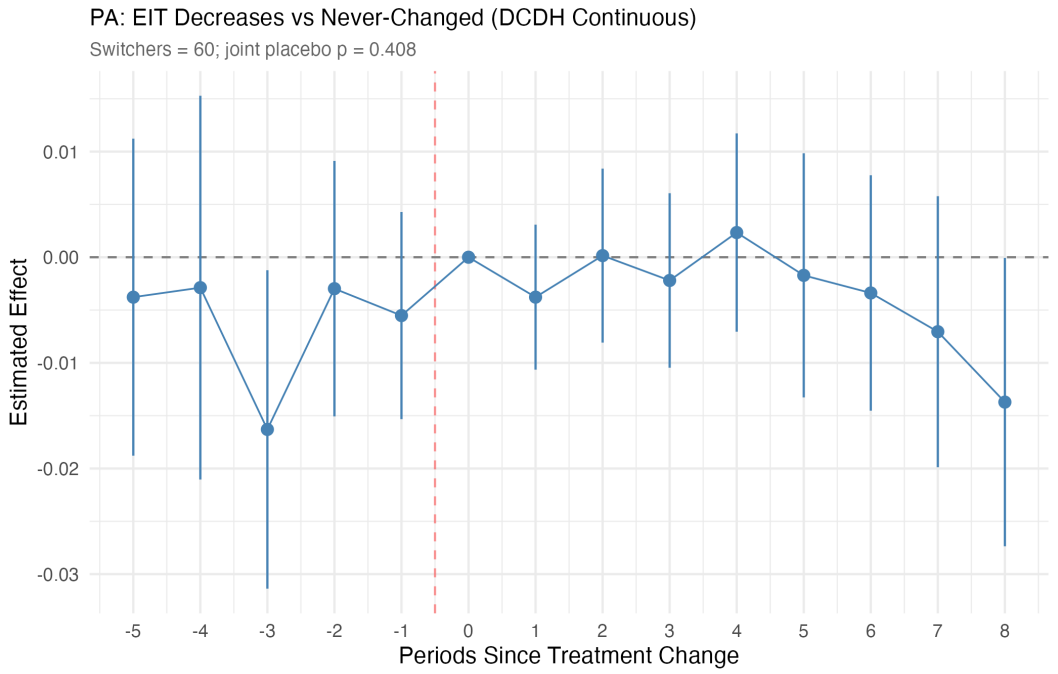


Figure 12: PA: DCDH Event Study — Continuous EIT Rate, Decreasers vs Never-Changed. 60 switchers. Rate decreases do not reverse residential sorting.

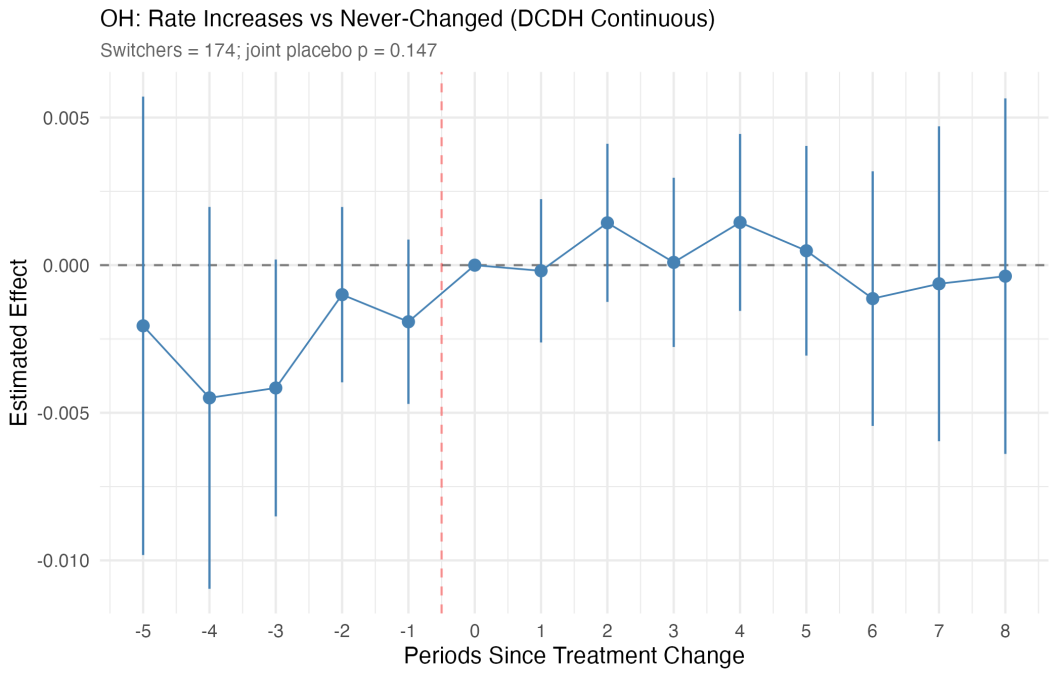


Figure 13: OH: DCDH Event Study — Continuous Municipal Rate, Increasers vs Never-Changed. 174 switchers. Estimates are imprecise due to the 0–3 rate scale.

5.3 Companion Outcomes: Decomposition

Table 7: Companion Outcomes: Decomposing the share_live_in Effect

	(1)	(2)	(3)	(4)
	share_live_in	log(Resident)	log(Nonresident)	log(Total Jobs)
PA Dose-Response (treated municipalities only)				
rate × post	−0.0099·	−0.1530*	−0.1012·	−0.1172*
	(0.0052)	(0.0720)	(0.0548)	(0.0549)
post	0.0022	0.1226·	0.1113·	0.1173*
	(0.0064)	(0.0721)	(0.0596)	(0.0587)
Diagnostics				
N	2,741	2,741	2,741	2,741
R^2	0.912	0.979	0.980	0.982
Within R^2	0.011	0.015	0.008	0.011
Fixed Effects		Municipality, Year		
Clusters		Municipality		
Equality test (2) vs (3)		$p = 0.09$		

Notes: Dose-response specification estimated among PA municipalities that raised EIT rates, 2002–2019. Column (1): main outcome. Columns (2)–(4): log counts of resident workers (live and work in municipality), nonresident commuters (work in municipality, live elsewhere), and total jobs. Equality test: joint model testing whether the resident coefficient differs from the nonresident coefficient. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

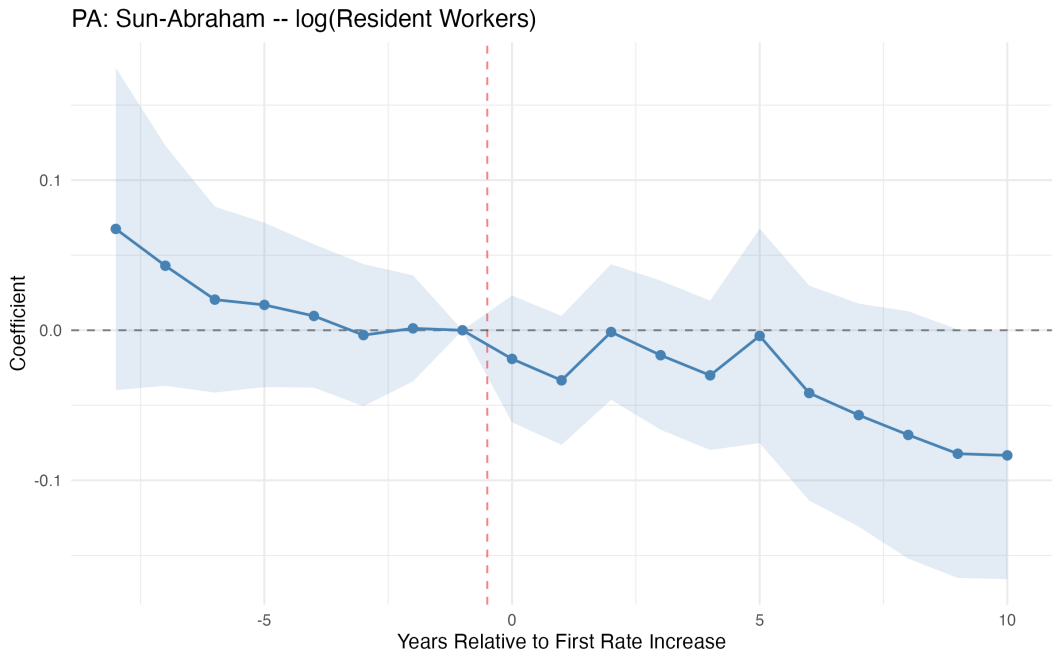


Figure 14: Pennsylvania: Sun-Abraham Event Study — log(Resident Workers).

5.4 Fiscal Impact

Table 8: Fiscal Impact of EIT Rate Changes (PA, 2002–2019)

	(1) log(EIT Rev)	(2) log(Total Tax)	(3) log(Total Rev)	(4) log(Police)	(5) log(Pop)	(6) log(Mkt Val)
resident_eit	0.6710**	0.2957**	0.0643	0.0531	−0.0094	0.0066
	(0.2106)	(0.0948)	(0.0429)	(0.0461)	(0.0137)	(0.0218)
N	42,309	42,526	42,543	22,979	42,523	9,323
R^2	0.977	0.983	0.959	0.953	0.997	0.978
Within R^2	0.027	0.006	0.0001	0.0001	0.0001	0.000
Fixed Effects	Municipality, Year					
Clusters	Municipality					

Notes: PA DCED Annual Financial Reports, 2002–2019. All outcomes in logs. Column (6) limited to 2015–2019 (market value data availability). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

5.5 Heterogeneity

6 Robustness

Table 9: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	−0.0126*	−0.0104*	−0.0165**			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				−0.0089**	−0.0079**	−0.0098**
				(0.0030)	(0.0028)	(0.0032)
N	27,277	14,174	27,257	3,058	2,173	3,038
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

6.1 Heterogeneity-Robust Continuous Treatment Estimator

As an additional robustness check, we implement the de Chaisemartin and D'Haultfoeuille (2024) estimator, which accommodates both continuous treatments and non-absorbing treatment paths under staggered timing. Unlike the Sun and Abraham (2021) estimator, which requires binary absorbing treatment, the DCDH estimator uses the actual tax rate as the treatment variable, allowing rate increases and decreases of varying magnitudes. This directly addresses the concern that our TWFE and Sun-Abraham specifications may suffer from heterogeneous treatment timing contamination when the treatment is continuous.

We estimate the DCDH specification separately for rate increases and decreases, each against never-changed controls. This mirrors the structure of our main Sun-Abraham results (Tables 2–3) while handling the continuous rate as treatment.

Table 10: Robustness: de Chaisemartin–D'Haultfoeuille (2024) Estimator

	Pennsylvania		Ohio	
	(1)	(2)	(3)	(4)
	Increases	Decreases	Increases	Decreases
Average Total Effect	−0.0078	−0.0077	0.0446	2.025
	(0.0069)	(0.0090)	(0.3132)	(1.398)
Effect at $\ell = 8$	-0.0111^{***}	—	−0.0023	—
	(0.0041)		(0.0030)	
Joint effects p	0.396	0.014	0.500	0.664
Joint placebo p	0.904	0.408	0.147	0.834
Switchers	120	60	174	22
Treatment	Continuous (EIT rate / municipal rate)			
Baseline trend	Linear polynomial in period-one treatment			

Notes: De Chaisemartin and D'Haultfoeuille (2024) interaction-weighted estimator with `continuous(1)`. Treatment is the municipal income tax rate (EIT in PA, municipal rate in OH). Columns (1)–(2): PA sample restricted to increasers (decreasers) plus never-changed municipalities. Columns (3)–(4): OH, same structure. Average Total Effect is the cumulative per-unit treatment effect averaged over ∼4.3 post-treatment periods. Joint placebo p tests the null that all pre-treatment coefficients are zero. Analytical standard errors clustered at the municipality level. OH continuous estimates are imprecise due to the 0–3 rate scale. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 10 reports the results. For Pennsylvania rate increases (column 1), the DCDH average total effect is −0.008 (SE 0.007), comparable to the Sun-Abraham ATT of −0.005 (SE 0.003) from Table 2. Pre-trends are clean (joint placebo $p = 0.90$), and the effect at

$\ell = 8$ reaches -0.011 ($p < 0.001$). The PA decrease estimate (column 2) shows a negative average effect (-0.008), reinforcing the asymmetry finding: rate decreases do not reverse sorting, consistent with sticky residential location decisions.

Ohio continuous estimates (columns 3–4) are uninformative due to the 0–3 rate scale, which inflates standard errors. The Ohio results rely on the binary Sun-Abraham estimator with no-tax controls (Table 3), where identification is sharper.

The consistency between the Sun-Abraham and DCDH estimates for Pennsylvania confirms that the main results are not driven by heterogeneous treatment timing contamination or the restriction to binary treatment.

7 Discussion

8 Conclusion

Acknowledgments

References

[1] Tiebout, C. M. (1956). A Pure Theory of Local Expenditures. *Journal of Political Economy*, 64(5), 416–424.

[2] Sun, L. & Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

[3] Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

A Pre-Treatment Balance

Table 11: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. All normalized differences < 0.15, within standard balance thresholds. Source: PA DCED Annual Financial Reports.

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on six fiscal observables from the PA DCED Annual Financial Reports using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 1 with municipality and year fixed effects, clustering standard errors at the municipality level. Table 12 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for five of six fiscal observables: total revenue (max $|t| = 1.60$), EIT revenue (1.33), streets expenditure (0.80), population (1.09), and total debt (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level.

Police expenditure is the exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase.

This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table 9), confirming that this pre-trend does not confound the estimated sorting response.

Table 12: Pre-Trend Tests: Sun-Abraham on Fiscal Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.60	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.33	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(Police Exp)	-0.085^{***}	-0.068^{**}	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.80	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the four pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure, where treated municipalities had significantly lower spending at $\ell = -5$ and $\ell = -4$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PA DCED Annual Financial Reports.

C Additional Figures